

3-Level Geometric Modeler: CSG, Subdivision Trees and Boundary Representation

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I. INTRODUCTION

Solid modeling is distinguished from related areas of geometric modeling and computer graphics by its emphasis on physical fidelity. Together, the principles of geometric and solid modeling form the foundation of computer-aided design and in general support the creation, exchange, visualization, animation, interrogation, and annotation of digital models of physical objects.

We will describe a polyhedron as three sets : (V, E, F) . V represents the set of vertices of the polyhedron. E represents the set of edges of the polyhedron. F represents the set of faces of the polyhedron.

II. DEFINITIONS

2.1 Regularized set

A set $S \subset \mathbb{R}^3$ is a regularized set iff $S = \overline{\text{int}(S)}$. All of our operations are on Regularized sets.

2.2 Regularized set operation on regularized sets A and B

- $A \cup^* B = \overline{\text{int}(A \cup B)}$
- $A \cap^* B = \overline{\text{int}(A \cap B)}$
- $A -^* B = \overline{\text{int}(A - B)}$
- $A^c * B = \overline{\text{int}(A^c)}$

2.3 Solid

A solid S is defined as a regularized subset of $\mathbb{R}^3$ ($S \subset \mathbb{R}^3$)

III. CSG OBJECTS

Just as a polyhedron can be described as the intersection of half spaces, a CSG (Constructive Solid Geometry) model describes a polyhedron as regularized set operation on primitives which are the simplest solid objects used for the representation.

3.1 Primitives

Primitives in CSG models are all semi-algebraic sets as follows:

1. Half Space: $H = \{(x, y, z) | a * x + b * y + c * z - d \geq 0\}$
2. Sphere : $S = \{(x, y, z) | (x - x_0)^2 + (y - y_0)^2 + (z - z_0)^2 - r^2 \leq 0\}$.

IV. SUBDIVISION TREES

1. Box: B is a box iff $B \subset \mathbb{R}^3$, $B = \{(x, y, z) | x_1 \leq x \leq x_2, y_1 \leq y \leq y_2, z_1 \leq z \leq z_2\}$, which is a regularized subset of $\mathbb{R}^3$
2. Elementary Box for a polyhedron P :
 - (a) Empty Box: a box B is an Empty Box iff $B \cap P = \emptyset$.

- (b) Full Box: a box B is a Full Box of Polyhedron P iff $B \cap P = B$
 - (c) Vertex Box: a box B is a Vertex Box iff there exists only one vertex $v \in V \cap B$, s.t. $\forall e \in E$, if $e \cap B \neq \emptyset$, v bounds e , and $\forall f \in F$, if $f \cap B \neq \emptyset$, v bounds f .
 - (d) Edge Box: a box B is an Edge Box iff $V \cap B = \emptyset$ and there exists only one edge $e \in E$, $e \cap B \neq \emptyset$, $\forall f \in F$, if $f \cap B \neq \emptyset$, e bounds f .
 - (e) Face Box: a box B is a Face Box iff $\forall e \in E$, $e \cap B = \emptyset$, and there exists one face $f \in F$, and $f \cap B \neq \emptyset$.
3. Subdivision tree : A subdivision tree in $\mathbb{R}^3$ is an octree, in which each node points to a box and we divide each box pointed by a inner node into 8 children pointing to 8 subboxes.
 4. Adequate subdivision tree for box B and polyhedron P $T = SD(B_0, P)$ in which SD is a function on a box B and a polyhedron P , the root of the tree points to box B_0 . All leaves of T point to an elementary box of the polyhedron P .
 5. Boolean operations on two adequate subdivision trees $T_1 = SD(B_1, P_1)$ and $T_2 = SD(B_2, P_2)$: $T = M(BO, T_1, T_2)$ in which B is a Boolean operation and T_1

and T_2 are adequate subdivision trees. The result of the function M is an adequate subdivision tree T .

if $B_1 \neq B_2$, $T = Null$.

if $B_1 = B_2 = B_0$, T is an adequate subdivision tree whose root points to B_0 . BO are Boolean operations as union ($\cup$), intersection ($\cap$) and difference ($-$).

We can also define:

- $T_1 \cup T_2 = M(\cup, T_1, T_2)$
- $T_1 \cap T_2 = M(\cap, T_1, T_2)$
- $T_1 - T_2 = M(-, T_1, T_2)$

6. Subdivision Rules of a box B_0 corresponding to a polyhedron P :

(a)

(b)

7. Boolean operation on an elementary box for two polyhedra P_1 and P_2
Suppose box B_e is an elementary box for both P_1 and P_2 . The the result of the Boolean operation on an elementary box for two polyhedra P_1 and P_2 is an adequate subdivision tree.

(a)